Algebraic Geometry Problem Set #5

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Problem (Exercise 1). *Let $n \geq 1$. Does there exist a surjective map $\mathbb{A}^n \to \mathbb{P}^n$?*

Remark. Further assume that the underlying field k is algebraically closed.

Solution. Yes, it exists. For starters, assume $n = 1$. Then, we construct the map $\mathbb{A}^1 \to \mathbb{P}^1$ by $x \mapsto [1 + x : x^2]$. It is clearly a well-defined morphism. To prove it is surjective, let us find the preimage of $[X_0 : X_1]$. If $x \in \mathbb{A}^1$ is the pre-image, then there exists $\lambda \in k^\times$ such that $1 + x = \lambda X_0$ and $x^2 = \lambda X_1$. Simplifying, this yields $X_0 x^2 - X_1 x - X_1 = 0$ which always has a solution over an algebraically closed k.

For general case $\mathbb{A}^n \to \mathbb{P}^n$, define

$$(x_1, \ldots, x_n) \mapsto [1 + x_1 + \cdots + x_n : x_1^2 : \cdots : x_n^2].$$

Let us find the preimage of $[X_0 : \cdots : X_n]$. For some $\lambda \in k^\times$, we must have

$$\begin{cases} x_i^2 = \lambda X_i, \ i = 1, \ldots, n \\ 1 + \sum_{i=1}^n x_i = \lambda X_0. \end{cases}$$

According to first n equations, we will search for a solution such that $x_i = \mu Z_i$ where $\mu^2 = \lambda$ and $Z_i^2 = X_i$. "Square roots" here make sense as k is algebraically closed. Substituting this into the last equation, we obtain

$$1 + \sum_{i=1}^n \mu Z_i = \mu^2 X_0 \Rightarrow X_0 \mu^2 - \left(\sum_{i=1}^n Z_i \right) \mu - 1 = 0,$$

which is a quadratic equation in μ. Note that $\mu = 0$ is definitely not the possible root of this equation. If $X_0 \neq 0$, this is a quadratic equation that over k has a solution. If $X_0 = 0$, we obtain the linear equation $\left(\sum_{i=1}^n Z_i\right) \mu = -1$. This equation has a solution iff $\sum_{i=1}^n Z_i \neq 0$. Note that we can always pick square roots Z_i to make this sum non-zero (note that all Z_i's cannot vanish to 0 simultaneously as $X_0 = 0$.)

Problem (Exercise 2). *Compute the zeta function $\zeta_{\mathbb{P}^n}(s)$ of the projective space $\mathbb{P}^n$ over a finite field $\mathbb{F}_q$.*

Solution. Recall that the zeta function is defined as

$$\zeta_X(T) := \exp\left(\sum_{m \geq 1} \frac{N_m}{m} T^m \right), \quad N_m = \#X(\mathbb{F}_{q^m}).$$

For $X = \mathbb{P}^n$, we have $\#\mathbb{P}^n(\mathbb{F}_{q^m}) = \sum_{j=0}^n q^{jm}$. Indeed, $\#\mathbb{P}^n(\mathbb{F}_{q^m})$ is the number of equivalence classes in $(\mathbb{F}_{q^m}^{n+1} \setminus 0)/\mathbb{F}_{q^m}^\times$, which is precisely $(q^{m(n+1)} - 1)/(q^m - 1)$.

Plug this result into the zeta function definition:

$$\begin{aligned}\zeta_{\mathbb{P}^n}(T) &= \exp\left(\sum_{m\geq 1}\sum_{j=0}^{n} q^{jm}\frac{T^m}{m}\right) = \exp\left(\sum_{j=0}^{n}\sum_{m\geq 1}\frac{(q^j)^m T^m}{m}\right)\\ &= \exp\left(\sum_{j=0}^{n}\left[-\log\left(1-q^jT\right)\right]\right) = \exp\left(-\log\left(\prod_{j=0}^{n}(1-q^jT)\right)\right)\\ &= \frac{1}{\prod_{j=0}^{n}(1-q^jT)}.\end{aligned}$$

Alternatively, by plugging in $T := q^{-s}$, we get $\zeta_{\mathbb{P}^n}(s) = 1/\prod_{j=0}^{n}(1-q^{j-s})$.

Problem (Exercise 3). *Compute $H^1(S^1, \underline{\mathbb{Z}})$ directly from our definition of cohomology. (Here S^1 is the circle.)*

Solution. Recall that the cohomology is defined through exact sequence

$$0 \to \Gamma(S^1, D^0(\underline{\mathbb{Z}})) \xrightarrow{d^0} \Gamma(S^1, D^1(\underline{\mathbb{Z}})) \xrightarrow{d^1} \Gamma(S^1, D^2(\underline{\mathbb{Z}}))$$

and $H^1(S^1, \underline{\mathbb{Z}}) = \ker(d^1)/\mathrm{im}(d^0)$. Also note that we have an exact sequence:

$$0 \to Q(\underline{\mathbb{Z}}) \to D^1(\underline{\mathbb{Z}}) \to D^2(\underline{\mathbb{Z}}),$$

which, thanks to right-exactness of $\Gamma(S^1, -)$, results in

$$0 \to \Gamma(S^1, Q(\underline{\mathbb{Z}})) \to \Gamma(S^1, D^1(\underline{\mathbb{Z}})) \xrightarrow{d^1} \Gamma(S^1, D^2(\underline{\mathbb{Z}})).$$

Thus we see that $\ker(d^1) \cong \Gamma(S^1, Q(\underline{\mathbb{Z}}))$ and so

$$H^1(S^1, \underline{\mathbb{Z}}) = \frac{\Gamma(S^1, Q(\underline{\mathbb{Z}}))}{\mathrm{im}[\Gamma(S^1, D^0(\underline{\mathbb{Z}})) \to \Gamma(S^1, Q(\underline{\mathbb{Z}}))]}$$

Now let us figure out what these global sections represent.

$\Gamma(S^1, D^0(\underline{\mathbb{Z}}))$. Note that $D^0(\underline{\mathbb{Z}})$ are arbitrary functions $S^1 \to \mathbb{Z}$, global section $\Gamma(S^1, D^0(\underline{\mathbb{Z}}))$ is an arbitrary function.

$\Gamma(S^1, Q(\underline{\mathbb{Z}}))$. Recall that $Q(\underline{\mathbb{Z}}) = D^0(\underline{\mathbb{Z}})/\underline{\mathbb{Z}}$. A global section $s \in \Gamma(S^1, Q(\underline{\mathbb{Z}}))$ is represented by a collection of local arbitrary functions $f_\alpha : U_\alpha \to \mathbb{Z}$ on an open cover $\{U_\alpha\}$. The rule is that $f_\alpha - f_\beta$ must be a locally constant integer-valued function.

To compute the quotient, we need to find sections on $Q(\underline{\mathbb{Z}})$ that cannot be lifted to a global section of $D^0(\underline{\mathbb{Z}})$. We do the standard procedure: cover S^1 with two intervals U_0 and U_1 overlapping in two disjoint smaller intervals $U_{01}^{\pm}$. Now we build the global section $s_n \in \Gamma(S^1, Q(\underline{\mathbb{Z}}))$ through two functions $f_U : U \to \mathbb{Z}$ and $f_V : V \to \mathbb{Z}$. Let $f_U \equiv 0$, $f_V|_{U_{01}^+} = 0$, $f_V|_{U_{10}^-} = n \in \mathbb{Z}$. The rest on f_V can be arbitrary.

When is s_n the image of $\Gamma(S^1, D^0(\underline{\mathbb{Z}}))$? It happens iff there is a single global function $F : S^1 \to \mathbb{Z}$ that maps down to s_n. Specifically, $F|_U = \alpha \in \mathbb{Z}$ and $F|_V = f_V + \beta$ for $\alpha, \beta \in \mathbb{Z}$. For F to be a well-defined global function, these formulas

must agree of overlaps, meaning $\alpha = \beta$ (for U_{10}^+) and $c_1 = n + c_2$ (on U_{10}^-). Combining those gives $n = 0$.

Thus, sections that cannot be lifted are determined by $n \in \mathbb{Z}$, which gives an isomorphism $H^1(S^1, \underline{\mathbb{Z}}) \cong \mathbb{Z}$.

Problem (Exercise 4). *Let X be a noetherian topological space and let $\{\mathcal{F}_\alpha\}$ be an inductive system of abelian sheaves on X. Show that the presheaf $U \mapsto \varinjlim \mathcal{F}_\alpha(U)$ is a sheaf, denoted by $\varinjlim \mathcal{F}_\alpha$. In particular, the natural map*

$$\varinjlim \Gamma(X, \mathcal{F}_\alpha) \to \Gamma(X, \varinjlim \mathcal{F}_\alpha)$$

is an isomorphism.

Solution. In what follows, let $\mathcal{F} := \varinjlim \mathcal{F}_\alpha$.

Let $U \subseteq X$ and let $\{U_i\}$ be an open cover. Suppose $s, t \in \mathcal{F}(U)$ have the same restrictions to each U_i. By definition of the colimit, there exist α and representatives $s_\alpha, t_\alpha \in \mathcal{F}_\alpha(U)$ mapping to s, t. Their restrictions agree in $\mathcal{F}(U_i)$ so for each i there exists $\beta_i \geq \alpha$ such that $s_\alpha|_{U_i} = t_\alpha|_{U_i}$ in $\mathcal{F}_{\beta_i}(U_i)$. Now use the fact that X is noetherian: every open subset is quasi-compact. We may assume that $U = \bigcup_{i=1}^n U_i$. Then, $s_\alpha, t_\alpha \mapsto s_\beta, t_\beta \in \mathcal{F}_\beta(U)$, and their restrictions agree on all U_i. Since $\mathcal{F}_\beta$ is a sheaf, $s_\beta = t_\beta$ hence $s = t$ in the colimit.

For gluing, let again $\{U_i\}$ be an open cover of U and suppose we are given compatible sections $s_i \in \mathcal{F}(U_i)$. Again, use the fact that the cover is finite, so choose $s_i \in \mathcal{F}_{\alpha_i}(U_i)$. Since the system is directed, there exists α dominating all α_i, so view all s_i as elements of $\mathcal{F}_\alpha(U_i)$. Compatibility means that for each i, j we have $s_i|_{U_{ij}} = s_j|_{U_{ij}}$. Thus for each pair i, j there exists $\beta_{ij} \geq \alpha$ such that they agree in $\mathcal{F}_{\beta_{ij}}(U_{ij})$. Since there are finitely many pairs, choose β dominating all β_{ij}. Then all s_i become sections in $\mathcal{F}_\beta(U_i)$ that agree on overlaps. Because $\mathcal{F}_\beta$ is a sheaf, there exists $s_\beta \in \mathcal{F}_\beta(U)$ such that $s_\beta|_{U_i} = s_i$. This defines a section $s \in \mathcal{F}(U)$, as needed.

Isomorphism is now quite evident. By definition of the colimit sheaf, $\Gamma(X, \varinjlim \mathcal{F}_\alpha) = (\varinjlim \mathcal{F}_\alpha)(X)$. Since our presheaf is a sheaf, $(\varinjlim \mathcal{F}_\alpha)(X) = \varinjlim \mathcal{F}_\alpha(X)$. Done.

Problem (Exercise 5). *Let D be a prime divisor on a smooth variety X. Show that $Pic(X \setminus D) \cong Pic(X)/\langle \mathcal{O}_X(D) \rangle$ where $\langle \mathcal{O}_X(D) \rangle = \{\mathcal{O}_X(nD) : n \in \mathbb{Z}\}$ is the subgroup generated by the class of $\mathcal{O}_X(D)$. Use this to compute $Pic(U)$ where $U = D(f) \subseteq \mathbb{P}^2$ is the complement of an irreducible curve of degree d.*

Solution. Start with the exact sequence

$$1 \to \mathcal{O}(X)^\times \to k(X)^\times \xrightarrow{\text{div}} \text{Div}(X) \to \text{Pic}(X) \to 1.$$

Let $U := X \setminus D$. Note that $k(U) = k(X)$, thus

$$k(X)^\times \xrightarrow{\text{div}_U} \text{Div}(U) \to \text{Pic}(U) \to 1.$$

Note that the restriction of a divisor $E = \sum a_i D_i$ is defined as $E|_U = \sum a_i (D_i \cap U)$. Since D is a prime divisor, D is the only prime divisor of X that does not intersect

U. Thus, $\mathrm{Div}(X) \to \mathrm{Div}(U)$ is surjective and its kernel is $\mathbb{Z}D$, thus

$$\mathrm{Div}(U) \cong \mathrm{Div}(X)/\mathbb{Z}D.$$

The restriction map of divisors induces a surjective morphism $\rho : \mathrm{Pic}(X) \to \mathrm{Pic}(U)$. A divisor $E \in \mathrm{Div}(X)$ is in the kernel of ρ iff its restriction $E|_U$ is a principal divisor on U, i.e. $E|_U = \mathrm{div}_U(g)$ for $g \in k(X)^\times$. Thus, in $\mathrm{Div}(X)$, we have $E = \mathrm{div}_X(g) + nD$ for some $n \in \mathbb{Z}$. Thus the class of E in $\mathrm{Pic}(X)$ is simply nD. Thus $\mathrm{Pic}(U) = \mathrm{Pic}(X)/\langle D \rangle$.

Now let us compute $\mathrm{Pic}(U)$ where $U = D(f) \subseteq \mathbb{P}^2$ is the complement of an irreducible curve of degree d. Let $X = \mathbb{P}^2$ and $D = V(f)$ so that $U = X \setminus D$. Recall that $\mathrm{Pic}(\mathbb{P}^2) \cong \mathbb{Z}$ and $\mathcal{O}_{\mathbb{P}^2}(D) \cong \mathcal{O}(d)$. Therefore,

$$\mathrm{Pic}(U) \cong \mathbb{Z}/\langle \mathcal{O}(d) \rangle = \mathbb{Z}/d\mathbb{Z}.$$